

Using Interior Information to Estimate Aggregate Effects in Ordered Multi-Threshold Regression Discontinuity Designs

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Abstract

Weighted averages of RD effects at ordered cutoffs decompose exactly into outer endpoint levels and smooth contrasts spanning adjacent cutoffs. Interior observations do not identify an individual RD jump, but they are informative about these segment contrasts. We estimate a contrast by integrating local-polynomial derivative estimates and derive its bias, variance, and limit distribution. We characterize its covariance with endpoint estimation and give a covariance-aware minimum-variance combination. Because the interior estimator converges more slowly under regular designs, the combination is first-order equivalent to endpoint estimation and any precision gain is higher-order. Matched-bandwidth simulations show modest finite-sample gains, particularly when endpoint information is relatively weak. The method is therefore a targeted augmentation for aggregate multi-threshold estimands, not a new source of causal identification or a claim of uniform efficiency dominance.

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1 Introduction

Regression discontinuity (RD) designs identify causal effects from discontinuities in conditional mean outcomes at known cutoffs (Hahn et al., 2001). In many applications, the running variable crosses more than one cutoff. Examples include age-based eligibility rules, school-entry rules repeated across cohorts, benefit schedules with several thresholds, and institutional rules that assign treatment intensity according to ordered score intervals. In such settings, researchers often summarize the evidence by estimating a weighted average of local discontinuities. Existing approaches typically estimate these effects by recentering observations around each cutoff, pooling the normalized running variable, or combining cutoff-specific estimates (Cattaneo et al., 2016; Bertanha, 2020). These methods are boundary-based: they use observations close to each cutoff to estimate the corresponding left and right limits of the conditional mean.

This paper studies a complementary source of information that arises in ordered multi-threshold designs. Let $t_1 < \dots < t_K$ denote known thresholds and let $g(z) = \mathbb{E}[Y \mid Z = z]$ be smooth on each open interval between adjacent thresholds, while allowing jumps at the thresholds. A weighted average of local discontinuities,

$$\tau = \sum_{k=1}^K \omega_k \{g(t_k^+) - g(t_k^-)\},$$

can be rewritten as a combination of outer endpoint terms and smooth contrasts between adjacent thresholds. These between-threshold contrasts are not RD jumps. They compare the right limit after one threshold, $g(t_{k-1}^+)$, with the left limit before the next threshold, $g(t_k^-)$. Under smoothness of g on (t_{k-1}, t_k) , the unweighted segment contrast satisfies

$$g(t_k^-) - g(t_{k-1}^+) = \int_{t_{k-1}}^{t_k} g'(z) dz.$$

Interior observations therefore do not identify a single local discontinuity, but

they do provide information about the smooth segment contrasts that enter aggregate multi-threshold estimands.

We use this decomposition to construct an interior estimator for each smooth segment contrast. For an adjacent-threshold segment (t_{k-1}, t_k) , define $\Delta_k = t_k - t_{k-1}$ and normalize $x = (z - t_{k-1})/\Delta_k \in [0, 1]$. Let

$$m_k(x) = g(t_{k-1} + x\Delta_k),$$

with endpoint limits inherited from the original scale, so that $m_k(0^+) = g(t_{k-1}^+)$ and $m_k(1^-) = g(t_k^-)$. The unweighted segment contrast is

$$m_k(1^-) - m_k(0^+) = \int_0^1 m'_k(u) du.$$

The proposed estimator partitions the normalized segment into bins, estimates $m'_k(u)$ at bin midpoints using local polynomial regression, and aggregates these derivative estimates by numerical integration. We derive its leading bias and asymptotic variance and establish a feasible central limit theorem.

The estimand distinction is the starting point of the paper. Interior observations cannot sharpen identification of an individual jump without additional causal restrictions. They can, however, estimate a smooth contrast that appears after several jumps are aggregated algebraically. The integrated-derivative estimator provides a second signal for this contrast by using variation throughout the segment.

The second step is information fusion. The interior and endpoint estimators are correlated in finite samples because both use observations in the first and last bins. Their correlation vanishes because the endpoint estimator is concentrated in shrinking boundary neighborhoods while the derivative estimator spreads weight across the segment. This first-order result does not, by itself, make the covariance irrelevant for the precision gain: that gain is of a smaller order under regular asymptotics. We therefore derive and recommend the covariance-aware minimum-variance combination. The familiar inverse-

variance rule is a diagonal approximation, useful only when the estimated covariance is negligible.

The paper makes three contributions. First, it gives an exact decomposition that separates causal jumps from smooth between-threshold contrasts and makes clear which pieces interior observations can estimate. Second, it develops an integrated-derivative estimator and derives its bias, variance, and feasible limit distribution. Third, it characterizes the boundary–interior covariance, supplies the appropriate minimum-variance combination, and shows explicitly that the augmentation does not improve the first-order convergence rate under a regular design. This last qualification turns the empirical question into the right one: when are the higher-order gains large enough to matter?

The paper contributes to the literature on RD designs and, more specifically, to work on designs with multiple cutoffs. Hahn et al. (2001) formalized identification and estimation in RD designs using discontinuities in conditional expectations. Subsequent work has refined local polynomial estimation and inference at RD boundaries, including robust bias-corrected inference and coverage-error-optimal bandwidth selection (Calonico et al., 2014, 2020). For multi-cutoff settings, Cattaneo et al. (2016) show how pooled and cutoff-specific estimands should be interpreted, Bertanha (2020) studies average treatment effects with many thresholds, and Cattaneo et al. (2021) use multiple cutoffs to extrapolate treatment effects under an additional identifying restriction. Our contribution is different: it does not extrapolate a causal effect away from a cutoff. It exploits the ordered structure of a prespecified aggregate estimand and uses interior observations only for the smooth segment contrasts in its algebraic decomposition.

The structure of the paper is as follows. Section 2 defines the ordered multi-threshold RD setting and derives the decomposition of aggregate estimands into boundary terms and smooth segment contrasts. Section 3 studies a single segment and develops boundary-level and integrated-derivative estimators for the same smooth contrast. Section 4 establishes their joint behavior and derives the covariance-aware combination. Section 5 extends the decomposition

and estimation strategy to weighted aggregate estimands. Section 6 presents simulation evidence under regular and sparse-boundary designs. Section 7 concludes.

2 Ordered Multi-Threshold RD Estimands

Empirical research often uses RD designs in settings where decisions depend on more than one cutoff. Examples include school-entry rules repeated across cohorts, age-based eligibility rules, benefit schedules with several thresholds, and institutional rules that assign treatment intensity according to ordered score intervals. In these settings, the running variable crosses a sequence of known thresholds, and each threshold may induce a discontinuity in the conditional mean outcome.

Consider N independent observations (z_i, y_i) , where $z_i \in [a, b]$. Let

$$g(z) = \mathbb{E}[y_i \mid z_i = z]$$

denote the conditional mean outcome. Suppose there are known ordered thresholds

$$t_1 < t_2 < \cdots < t_K,$$

and that $g(\cdot)$ is smooth on each open interval between adjacent thresholds, while allowing jumps at the thresholds:

$$g(t_k^-) \neq g(t_k^+), \quad k = 1, \dots, K.$$

The local discontinuity at threshold t_k is

$$\tau_k = g(t_k^+) - g(t_k^-).$$

Researchers often summarize local discontinuities by a weighted average,

$$\tau = \sum_{k=1}^K \omega_k \tau_k, \quad \text{with } \sum_{k=1}^K \omega_k = 1.$$

The weights ω_k may be chosen by the researcher, set equal across thresholds, or induced by the empirical distribution of observations near the thresholds. Standard multi-cutoff RD approaches typically estimate such objects by recentering observations around each cutoff, pooling the normalized running variable, or combining cutoff-specific estimates (Cattaneo et al., 2016; Bertanha, 2020).

The ordered structure of the thresholds implies

$$\tau = \omega_K g(t_K^+) - \omega_1 g(t_1^-) - \sum_{k=2}^K [\omega_k g(t_k^-) - \omega_{k-1} g(t_{k-1}^+)].$$

Equivalently,

$$\tau = \omega_K g(t_K^+) - \omega_1 g(t_1^-) - \sum_{k=2}^K \delta_k, \tag{1}$$

where

$$\delta_k = \omega_k g(t_k^-) - \omega_{k-1} g(t_{k-1}^+).$$

The terms τ_k are RD jumps. In contrast, the terms δ_k are weighted between-threshold contrasts. When adjacent weights are equal, $\omega_k = \omega_{k-1}$, δ_k is proportional to the smooth segment contrast $g(t_k^-) - g(t_{k-1}^+)$. With unequal adjacent weights, δ_k also contains the difference in weights attached to the two endpoint levels. Section 5 treats this general weighted case explicitly. The key point is that, when $g(\cdot)$ is smooth on (t_{k-1}, t_k) , interior observations are informative about the smooth component of these between-threshold terms. They do not identify a single RD jump, but they can identify segment contrasts that enter aggregate parameters such as (1).

For the main theoretical analysis, we study one such segment after normal-

ization. On the segment $[t_{k-1}, t_k]$, let $\Delta_k = t_k - t_{k-1} > 0$ and define

$$x = (z - t_{k-1}) / \Delta_k \in [0, 1], \quad m(x) = g(t_{k-1} + x\Delta_k).$$

Then

$$m(0^+) = g(t_{k-1}^+), \quad m(1^-) = g(t_k^-),$$

and

$$m'(x) = \Delta_k g'(t_{k-1} + x\Delta_k).$$

The unweighted segment contrast is

$$\delta = m(1^-) - m(0^+) = \int_0^1 m'(u) du. \quad (2)$$

The weighted case in (1) is considered in Section 5.

3 Boundary and Interior Estimation of a Segment Contrast

This section studies estimation of the smooth segment contrast δ in (2). We observe independent data $\{(X_i, Y_i)\}_{i=1}^N$ satisfying

$$Y_i = m(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | X_i] = 0, \quad \text{Var}(\varepsilon_i | X_i) = \sigma^2(X_i),$$

where $X_i \in [0, 1]$. The target is the difference between the two one-sided endpoint limits,

$$\delta = m(1^-) - m(0^+).$$

There are two signals for this contrast. The boundary-level estimator estimates $m(0^+)$ and $m(1^-)$ directly using one-sided local polynomial regressions. The integrated-derivative estimator uses the identity $\delta = \int_0^1 m'(u) du$ and estimates m' inside the segment.

3.1 Boundary-level estimator

The boundary-level estimator is

$$\hat{\delta}_{\text{levels}} = \hat{m}(1^-) - \hat{m}(0^+).$$

This is the standard endpoint estimator for a smooth contrast. It is analogous to the level-estimation step used in RD estimation, except that the target here is not a jump at a cutoff but a contrast between the right limit after one threshold and the left limit before the next threshold.

Let $\hat{m}(0^+)$ and $\hat{m}(1^-)$ denote the intercepts from one-sided degree- p local polynomial regressions at the left and right endpoints. For example, at the left endpoint,

$$\hat{\alpha}_0 = \arg \min_{\alpha} \sum_{i=1}^N \left\{ Y_i - \sum_{j=0}^p \alpha_j X_i^j \right\}^2 K(X_i/h_0) \mathbf{1}\{0 \leq X_i \leq h_0\}, \quad \hat{m}(0^+) = e_1^\top \hat{\alpha}_0,$$

where $e_1 = (1, 0, \dots, 0)^\top$. The right endpoint estimator is defined analogously using observations with $1 - h_1 \leq X_i \leq 1$, centered at 1. The selector is e_1 , not the derivative selector, because the boundary estimator targets endpoint levels.

From standard local polynomial theory (Fan and Yao, 2005), if $h_0 = c_0 h$, $h_1 = c_1 h$, and $h \rightarrow 0$, then

$$\sqrt{Nh} [\hat{\delta}_{\text{levels}} - \delta - \text{Bias}_{\text{levels}}] \xrightarrow{d} \mathcal{N} \left(0, \frac{R_1}{c_1} \frac{\sigma^2(1^-)}{f(1^-)} + \frac{R_0}{c_0} \frac{\sigma^2(0^+)}{f(0^+)} \right), \quad (3)$$

where R_0 and R_1 are the squared one-sided equivalent-kernel constants for the endpoint intercept estimators. The leading bias $\text{Bias}_{\text{levels}}$ is the usual one-sided local-polynomial endpoint bias of order h^{p+1} . In the simulations below, $p = 1$, and the endpoint intercepts are estimated by one-sided local linear regressions with the triangular kernel.

Under regular designs, the boundary-level estimator has the familiar $\sqrt{Nh}$ convergence rate. It is the benchmark estimator for the segment contrast. The

integrated-derivative estimator introduced next provides an additional signal for the same contrast.

3.2 Integrated-derivative estimator

The integrated-derivative estimator uses

$$\delta = m(1^-) - m(0^+) = \int_0^1 m'(u) du.$$

Partition the interval $[0, 1]$ into B subintervals of length $2h$, with $h = 1/(2B)$, and let c_b denote the midpoint of bin b . At each midpoint c_b , estimate $m'(c_b)$ by local polynomial regression using observations in

$$I_b = [c_b - h, c_b + h].$$

Specifically,

$$\hat{\beta}(c_b) = \arg \min_{\beta} \sum_{X_i \in I_b} \left(Y_i - \sum_{j=0}^p \beta_j (X_i - c_b)^j \right)^2 K_h(X_i - c_b),$$

with $K_h(u) = K(u/h)/h$. The estimator of $m'(c_b)$ is

$$\hat{\beta}_1(c_b) = e_2^T \hat{\beta}(c_b) = \sum_{i=1}^N w_{i,b} Y_i, \quad (4)$$

where $w_{i,b} = w^T((X_i - c_b)/h)$, and

$$w^T(u) = e_2^T S_N^{-1} [1, uh, \dots, (uh)^p]^T \frac{K(u)}{h}.$$

The matrix S_N has (i, j) th element $S_{N,i+j-2}$, with

$$S_{N,j} = \sum_{i=1}^N K_h(X_i - c_b) (X_i - c_b)^j.$$

Aggregating the derivative estimates gives

$$\hat{\beta}_{\text{avg}} = 2h \sum_{b=1}^B \hat{\beta}_1(c_b) = \sum_{i=1}^N w_{N,i} Y_i,$$

where $w_{N,i}$ is the total weight assigned to observation i .

We impose the following assumptions.

Assumption 1 (Smoothness of m). $m \in C^{\max\{p+1, 5\}}([0, 1])$ for some integer $p \geq 1$.

Assumption 2 (Design Density). Each X_i takes values in $[0, 1]$ with a density f that is continuous, strictly positive on $[0, 1]$ (no atoms), and has a bounded second derivative.

Assumption 3 (Bounded Conditional Variance). There exist constants $0 < \underline{\sigma}^2 \leq \bar{\sigma}^2 < \infty$ such that

$$\underline{\sigma}^2 \leq \sigma^2(x) = \text{Var}(\varepsilon_i \mid X_i = x) \leq \bar{\sigma}^2$$

for all $x \in [0, 1]$, and $\sigma^2(\cdot)$ is continuous.

Assumption 4 (Kernel Conditions). The kernel K is symmetric, bounded, supported on $[-1, 1]$, and satisfies

$$\int_{-1}^1 K(u) du = 1, \quad \int_{-1}^1 |u|^{p+1} |K(u)| du < \infty.$$

Define the equivalent kernel for the derivative estimator by

$$K^*(u) = e_2^T S^{-1} (1, u, \dots, u^p)^T K(u),$$

where S is a $(p+1) \times (p+1)$ matrix with entries

$$S_{i,j} = \mu_{i+j-2} \quad \text{and} \quad \mu_j = \int u^j K(u) du.$$

It follows that K^* is also supported and bounded on $[-1, 1]$.

Assumption 5 (Bandwidth). Let $h = h(N) \rightarrow 0$ as $N \rightarrow \infty$ and assume

$$\frac{Nh}{\log N} \rightarrow \infty, \quad Nh^2 \rightarrow \infty, \quad \sqrt{N}h^3 \rightarrow 0.$$

Assumption 6 (Higher-Order Moment of ε_i). There exists $\eta > 0$ and a constant $C < \infty$ such that

$$\mathbb{E}[|\varepsilon_i|^{2+\eta} \mid X_i = x] \leq C \quad \text{for all } x \in [0, 1],$$

uniformly in i .

4 Combining Boundary and Interior Signals

This section gives the main information-fusion result. The boundary-level estimator and the integrated-derivative estimator both target

$$\delta = m(1^-) - m(0^+).$$

They use different parts of the segment. Both estimators are valid estimators of the same smooth contrast. The distinction is not identification but information: one signal is concentrated at the endpoints, while the other is distributed across the segment. We show that their correlation vanishes and derive their covariance from their effective weights. That covariance is retained in the recommended combination because it can matter at the same order as the finite-sample precision gain.

4.1 Asymptotic distribution of the integrated-derivative estimator

Theorem 1 (Integrated-derivative estimator for a segment contrast). *Let $\{(X_i, Y_i)\}_{i=1}^N$ be independent observations from the model*

$$Y_i = m(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | X_i] = 0, \quad \text{Var}(\varepsilon_i | X_i) = \sigma^2(X_i),$$

where $X_i \in [0, 1]$ has density f . Suppose Assumptions 1–6 hold with polynomial degree $p \geq 2$, so that $m(\cdot)$ is $(p+1)$ -times continuously differentiable on $[0, 1]$, and we use a degree- p local polynomial estimator to estimate $m'(\cdot)$. Define

$$\Gamma = \int_0^1 m'(u) \, du = m(1^-) - m(0^+).$$

Partition $[0, 1]$ into B subintervals each of length $2h = 1/B$, with midpoints $c_b = (2b-1)h$ for $b = 1, \dots, B$. For each midpoint c_b , let $\hat{\beta}_1(c_b)$ be the local polynomial estimator of the derivative $m'(c_b)$. Define

$$\hat{\beta}_{\text{avg}} = 2h \sum_{b=1}^B \hat{\beta}_1(c_b).$$

Then, under Assumptions 1–6, and conditional on the design $X = \{X_1, \dots, X_N\}$,

$$\sqrt{N}h^2 \left(\hat{\beta}_{\text{avg}} - \Gamma - \text{Bias} \right) \xrightarrow{d} \mathcal{N} \left(0, 2 \int_{-1}^1 [K^*(u)]^2 \, du \int_0^1 \frac{\sigma^2(u)}{f(u)} \, du \right),$$

where

- If $p > 2$, then the local-polynomial bias is $o(h^2)$, and the dominant bias term comes from the midpoint rule:

$$\text{Bias} = -\frac{h^2}{6} \left[m''(1^-) - m''(0^+) \right].$$

- If $p = 2$, both the midpoint rule and the local-polynomial estimator contribute

to the bias at order h^2 :

$$\text{Bias} = \frac{h^2}{6} \left(\int_{-1}^1 u^3 K^*(u) du - 1 \right) [m''(1^-) - m''(0^+)].$$

Theorem 1 establishes that the interior estimator is asymptotically normal and has a feasible variance expression. Its variance depends on the density and conditional variance over the segment through

$$\int_0^1 \frac{\sigma^2(u)}{f(u)} du.$$

This differs from the boundary-level estimator in (3), whose asymptotic variance depends on the behavior of f and σ^2 at 0^+ and 1^- .

4.2 Vanishing correlation and joint limit

Theorem 2 (Vanishing boundary–interior correlation). *Let $\hat{\beta}_{\text{avg}}$ be the integrated-derivative estimator with bandwidth h as in Theorem 1, and write*

$$\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} | X] = \sum_{i=1}^N w_i \varepsilon_i, \quad \max_i |w_i| \leq \frac{C}{Nh}, \quad \sum_{i=1}^N w_i^2 = \Theta_p\left(\frac{1}{Nh^2}\right).$$

Let the boundary levels at 0 and 1 be estimated by one-sided degree- p local polynomials with asymmetric equivalent kernels and bandwidths $h_0 = c_0 h$ (at 0) and $h_1 = c_1 h$ (at 1), where $0 < c_{\min} \leq c_0, c_1 \leq c_{\max} < \infty$:

$$\hat{\delta}_{\text{levels}} = \hat{m}(1^-) - \hat{m}(0^+), \quad \hat{\delta}_{\text{levels}} - \mathbb{E}[\hat{\delta}_{\text{levels}} | X] = \sum_{i=1}^N v_i \varepsilon_i,$$

with

$$\max_i |v_i| \leq \frac{C}{Nh}, \quad \sum_{i=1}^N v_i^2 = \Theta_p\left(\frac{1}{Nh}\right),$$

and $v_i \neq 0$ only if $X_i \in [0, A_0 h] \cup [1 - A_1 h, 1]$, where $A_0, A_1 > 0$ depend only on the one-sided equivalent kernels and on c_0, c_1 .

Under Assumptions 1–6 (in particular $Nh \rightarrow \infty$ and $h \rightarrow 0$ and Assumption 2),

conditionally on X ,

$$\text{Cov}(\hat{\beta}_{\text{avg}}, \hat{\delta}_{\text{levels}} \mid X) = O_p\left(\frac{A_0 f(0+) + A_1 f(1-)}{Nh}\right),$$

and

$$\text{Corr}(\hat{\beta}_{\text{avg}}, \hat{\delta}_{\text{levels}} \mid X) = O_p\left((A_0 f(0+) + A_1 f(1-)) h^{1/2}\right) \rightarrow 0.$$

In particular, a Cramér–Wold argument for the two linear representations yields a jointly Gaussian limit with diagonal covariance after each estimator is standardized by its own asymptotic standard deviation.

The proof is based on the covariance of two linear representations. Conditional on the design,

$$\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X] = \sum_{i=1}^N w_i \varepsilon_i, \quad \hat{\delta}_{\text{levels}} - \mathbb{E}[\hat{\delta}_{\text{levels}} \mid X] = \sum_{i=1}^N v_i \varepsilon_i.$$

Thus,

$$\text{Cov}(\hat{\beta}_{\text{avg}}, \hat{\delta}_{\text{levels}} \mid X) = \sum_{i=1}^N w_i v_i \sigma^2(X_i).$$

The boundary weights v_i are nonzero only when X_i lies in endpoint neighborhoods of total length proportional to h . The number of observations in these neighborhoods is therefore $O_p(Nh)$. Since $\max_i |w_i| \leq C/(Nh)$ and $\max_i |v_i| \leq C/(Nh)$, the covariance is $O_p((Nh)^{-1})$. Dividing by the product of the standard deviations, of orders $(Nh^2)^{-1/2}$ and $(Nh)^{-1/2}$, gives a correlation of order $O_p(h^{1/2})$, which converges to zero.

4.3 Covariance-aware combination

Let

$$\hat{\delta}_{\text{int}} = \hat{\beta}_{\text{avg}}$$

denote the integrated-derivative estimator and let

$$\hat{\delta}_{\text{bnd}} = \hat{\delta}_{\text{levels}}$$

denote the boundary-level estimator. Let $\hat{V}_{\text{int}}$ and $\hat{V}_{\text{bnd}}$ denote feasible variance estimators for $\hat{\delta}_{\text{int}}$ and $\hat{\delta}_{\text{bnd}}$, respectively, on the original estimator scale, and let $\hat{C}$ estimate their covariance. Consider

$$\hat{\delta}_{\text{comb}} = \lambda \hat{\delta}_{\text{bnd}} + (1 - \lambda) \hat{\delta}_{\text{int}}, \quad \lambda = \frac{\hat{V}_{\text{int}} - \hat{C}}{\hat{V}_{\text{bnd}} + \hat{V}_{\text{int}} - 2\hat{C}}.$$

This is the minimum-variance linear combination of two estimators with the same target. Its feasible variance is

$$\hat{V}_{\text{comb}} = \frac{\hat{V}_{\text{bnd}} \hat{V}_{\text{int}} - \hat{C}^2}{\hat{V}_{\text{bnd}} + \hat{V}_{\text{int}} - 2\hat{C}}.$$

Because both estimators are linear smoothers conditional on the running variables, a direct covariance estimator is available. If w_i and v_i are their effective weights, then

$$\hat{C} = \sum_{i=1}^N w_i v_i \hat{\sigma}_i^2,$$

where $\hat{\sigma}_i^2$ is a locally estimated conditional error variance. If $\hat{C}$ is set to zero, the formula reduces to the usual inverse-variance combination. The zero-covariance version is a convenient diagonal approximation, but the vanishing-correlation theorem alone does not justify it for a higher-order efficiency comparison: the covariance contribution can be of the same order as the gain relative to the boundary estimator. For numerical stability, an implementation may truncate λ to $[0, 1]$, at the cost of replacing the unconstrained minimum-variance rule by its convex-combination analogue.

Under regular designs, $V_{\text{bnd}} = O((Nh)^{-1})$ while $V_{\text{int}} = O((Nh^2)^{-1})$. Hence $1 - \lambda = O_p(h)$, and the combined estimator is first-order equivalent to the boundary estimator. The method does not improve its first-order rate or first-

order variance. It is most useful as a finite-sample augmentation when the density is low near 0^+ or 1^- , the conditional variance is high near the endpoints, or ordered multi-threshold designs produce thin local samples around some thresholds.

5 Weighted Segment Contrasts and Multi-Threshold Aggregation

For a segment with endpoint levels $a = m(0^+)$ and $b = m(1^-)$, the aggregate decomposition contains

$$\delta_w = \omega_1 b - \omega_0 a.$$

Let $\bar{\omega} = (\omega_1 + \omega_0)/2$ and $\Delta\omega = \omega_1 - \omega_0$. Then

$$\delta_w = \bar{\omega}(b - a) + \frac{\Delta\omega}{2}(b + a). \quad (5)$$

The first component is the smooth segment difference and can be estimated by either the boundary estimator, the integrated-derivative estimator, or their covariance-aware combination. The second is an endpoint-level component; interior derivatives do not identify it. It must be estimated from the two endpoint levels or from an explicitly stated model for levels. Thus, interior data can augment the entire segment term when adjacent aggregation weights are equal, but only its common-weight component when the weights differ.

Let $\hat{d}_{\text{comb}}$ estimate $b - a$, and let $\hat{s}_{\text{bnd}} = \hat{b} + \hat{a}$. A transparent estimator is

$$\hat{\delta}_w = \bar{\omega}\hat{d}_{\text{comb}} + \frac{\Delta\omega}{2}\hat{s}_{\text{bnd}}. \quad (6)$$

Its variance must include the covariance between $\hat{d}_{\text{comb}}$ and $\hat{s}_{\text{bnd}}$, available from their effective linear-smoother weights. For the aggregate parameter in (1), apply (6) to each adjacent segment and assemble the variance from the effective weight of the full estimator. This automatically accounts for any ob-

servations shared by neighboring segment and outer-endpoint estimators.

6 Simulation Evidence

6.1 Single-segment evidence

We first examine the normalized single-segment problem. We compare three estimators of the smooth segment contrast

$$\delta = m(1^-) - m(0^+).$$

The first is a boundary-level estimator,

$$\hat{\delta}_{\text{bnd}} = \hat{m}(1^-) - \hat{m}(0^+),$$

computed using one-sided local linear regressions at 0^+ and 1^- with the triangular kernel $K(u) = \max\{0, 1 - |u|\}$. The second is the integrated-derivative estimator,

$$\hat{\delta}_{\text{int}} = \sum_{b=1}^B \ell_B \hat{\beta}_1(c_b),$$

which estimates $m'(c_b)$ at bin midpoints using local quadratic regressions and aggregates the slope estimates by midpoint quadrature. The third is the covariance-aware combined estimator,

$$\hat{\delta}_{\text{comb}} = w_{\text{bnd}} \hat{\delta}_{\text{bnd}} + w_{\text{int}} \hat{\delta}_{\text{int}},$$

with weights computed from feasible variance and covariance estimates as in Section 4.

We partition $[0, 1]$ into B equal bins of width $\ell_B = 1/B$ and set $h_0 = \ell_B/2$. The number of bins is chosen as $B(n) = \max\{4, \text{round}(1.5n^{1/4})\}$, so $B = 10$ when $n = 2000$. For bin b , we run a local quadratic regression of Y on $(1, X - c_b, (X - c_b)^2)$ with triangular weights on $|X - c_b| \leq h_0$, and take the slope

coefficient $\hat{\beta}_1(c_b)$ as the estimate of $m'(c_b)$.

The implementation extracts the exact linear-smoother weights from every weighted least-squares fit. If w_i and v_i denote the effective integrated and boundary weights, respectively, locally weighted residual variances give

$$\hat{V}_{\text{int}} = \sum_i w_i^2 \hat{\sigma}_i^2, \quad \hat{V}_{\text{bnd}} = \sum_i v_i^2 \hat{\sigma}_i^2, \quad \hat{C} = \sum_i w_i v_i \hat{\sigma}_i^2.$$

We use the covariance-aware weight in Section 4, truncated to $[0, 1]$. The simulation code also computes the zero-covariance diagonal rule and, for the single-segment experiment, an oracle rule using the known conditional error variance.

Remark 1 (Implementation bandwidth). The rule $B(n) = \max\{4, \text{round}(1.5n^{1/4})\}$ is a feasible implementation rule used in the simulations, not an optimal bandwidth rule. Since $h_0 = \ell_B/2 = 1/(2B)$, this choice gives $h_0 \asymp n^{-1/4}$, which satisfies the bandwidth restrictions in Assumption 5. In the main comparison, the endpoint and derivative fits use the same numerical bandwidth h_0 . This matched-bandwidth design holds the local neighborhood fixed and isolates the effect of adding the interior signal. It does not claim that h_0 is separately MSE-optimal for either estimator. Deriving a joint, method-specific bandwidth selector is beyond the scope of the present paper; applications should report sensitivity to the bin and bandwidth choice.

We draw X_i either from a uniform distribution on $[0, 1]$ or from a truncated normal distribution $\mathcal{N}(0.5, 0.3^2)$ on $[0, 1]$. The latter design places less mass near the endpoints and therefore weakens the boundary-level estimator. Outcomes are generated as

$$Y_i = m(X_i) + \varepsilon_i,$$

with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2(X_i))$. We consider both homoskedastic errors, $\sigma(x) = 1$, and heteroskedastic errors, $\sigma^2(x) = 1 + x$. We use $n_{\text{sim}} = 500$ Monte Carlo replications and report root mean squared error, the RMSE gain relative to boundary estimation, and empirical coverage of the feasible 95% Wald interval.

We consider two conditional mean functions:

$$m(x) = 2x + x^2, \quad m(x) = \frac{1 - \cos(\pi x)}{2}.$$

For the polynomial design, $\delta = 3$. For the cosine design, $\delta = 1$. The replication code is available at https://github.com/vahid-moghani89/Difference_Estimation.

Scenario	Bnd. RMSE	Int. RMSE	Comb. RMSE	Gain (%)	Cov.95	Bnd. weight	Corr.
<i>Panel A: Polynomial DGP $m(x) = 2x + x^2$</i>							
Uniform – Homo, $n = 2000$	0.301	0.961	0.294	2.6	0.954	0.936	0.078
Uniform – Het., $n = 2000$	0.365	1.186	0.352	3.4	0.946	0.932	0.064
Trunc. normal – Homo, $n = 2000$	0.456	1.074	0.437	4.3	0.940	0.876	0.109
Trunc. normal – Het., $n = 2000$	0.644	1.349	0.616	4.3	0.942	0.846	0.129
Uniform – Homo, $n = 1000$	0.374	1.131	0.360	3.7	0.940	0.924	0.071
Uniform – Homo, $n = 4000$	0.248	0.807	0.241	2.9	0.950	0.933	0.069
<i>Panel B: Cosine DGP $m(x) = \frac{1 - \cos(\pi x)}{2}$</i>							
Uniform – Homo, $n = 2000$	0.315	0.955	0.308	2.4	0.936	0.930	0.073
Uniform – Het., $n = 2000$	0.356	1.188	0.350	1.7	0.956	0.938	0.089
Trunc. normal – Homo, $n = 2000$	0.560	1.095	0.524	6.4	0.944	0.830	0.109
Trunc. normal – Het., $n = 2000$	0.683	1.288	0.627	8.2	0.938	0.841	0.128
Uniform – Homo, $n = 1000$	0.384	1.133	0.377	1.8	0.956	0.914	0.080
Uniform – Homo, $n = 4000$	0.275	0.852	0.262	4.8	0.956	0.922	0.075

Notes: The table uses 500 replications. Comb. is the feasible covariance-aware convex combination. Gain is its percentage RMSE reduction relative to Bnd. Cov.95 is coverage of its feasible Wald interval. Bnd. weight is the average λ ; Corr. is the average estimated boundary–interior correlation. The diagonal and oracle results are reported in the replication files.

Table 1: Single-segment covariance-aware simulation results

The covariance-aware estimator reduces RMSE relative to the boundary estimator in all 12 designs. The median gain is 3.6%, ranging from 1.7% to 8.2%. Feasible weights closely track the oracle weights, and coverage ranges from 93.6% to 95.6%. The combined estimator still places 83–94% weight on the boundary signal, consistent with its faster first-order rate.

6.2 Ordered multi-threshold evidence

We simulate ordered multi-threshold RD designs with $K \in \{5, 10\}$ cutoffs at

$$t_k = \frac{k}{K+1}, \quad k = 1, \dots, K.$$

We use $n = 2000K$, so $n = 10,000$ for $K = 5$ and $n = 20,000$ for $K = 10$. The target is the equal-weight average of the local discontinuities. Outcomes are generated from

$$Y_i = q(Z_i) + \sum_{k=1}^K \tau_k \mathbf{1}\{Z_i \geq t_k\} + \varepsilon_i,$$

where the heterogeneous jumps are generated by

$$\begin{aligned} \tau_k &= 0.6 + 0.25(t_k - 0.5) \\ &+ 0.60 \left\{ (t_k - 0.5)^2 - \frac{1}{K} \sum_{j=1}^K (t_j - 0.5)^2 \right\} + 0.15 \sin\left(\frac{2\pi k}{K}\right). \end{aligned}$$

Thus $K^{-1} \sum_k \tau_k = 0.6$. The resulting vectors are

$$\boldsymbol{\tau}_5 = (0.692658, 0.629834, 0.478499, 0.482342, 0.716667)$$

and

$$\begin{aligned} \boldsymbol{\tau}_{10} &= (0.645399, 0.682948, 0.675923, 0.624325, 0.548967, \\ &0.483526, 0.461680, 0.504242, 0.611212, 0.761777). \end{aligned}$$

We consider two smooth baselines,

$$q(z) = 2z + z^2, \quad q(z) = \frac{1 - \cos(\pi z)}{2},$$

and draw Z_i either uniformly on $[0, 1]$ or from $\mathcal{N}(0.5, 0.3^2)$ truncated to $[0, 1]$.

Errors are either homoskedastic, $\sigma(z) = 1$, or heteroskedastic, $\sigma^2(z) = 1 + z$.

Each design uses 300 Monte Carlo replications.

We compare two estimators. The first estimates each cutoff-specific RD jump using one-sided local linear regressions and averages the estimates with equal weights. The second uses the decomposition in (1): the two outer end-point terms are estimated by one-sided local linear regressions, while each between-threshold segment contrast is estimated by the covariance-aware combination of its boundary-level and integrated-derivative estimators. Each segment is normalized to $[0, 1]$ before applying the integrated-derivative estima-

tor, with $B(n) = \max\{4, \text{round}(1.5n^{1/4})\}$ bins. The effective weights of the full decomposition estimator are used for its feasible standard error.

Baseline	Design	K	Errors	Sep. RMSE	Comb. RMSE	Gain (%)	Cov.95	Bnd. weight
Poly.	Uniform	5	Homo	0.148	0.144	2.8	0.963	0.928
Poly.	Uniform	10	Homo	0.108	0.107	0.6	0.933	0.928
Poly.	Uniform	5	Het.	0.197	0.191	3.0	0.943	0.928
Poly.	Uniform	10	Het.	0.122	0.118	3.0	0.950	0.929
Poly.	Trunc. normal	5	Homo	0.148	0.144	2.9	0.953	0.927
Poly.	Trunc. normal	10	Homo	0.109	0.106	2.8	0.947	0.929
Poly.	Trunc. normal	5	Het.	0.188	0.182	2.9	0.943	0.927
Poly.	Trunc. normal	10	Het.	0.133	0.131	1.4	0.947	0.928
Cos.	Uniform	5	Homo	0.156	0.150	3.9	0.947	0.928
Cos.	Uniform	10	Homo	0.109	0.104	3.8	0.950	0.928
Cos.	Uniform	5	Het.	0.193	0.191	0.7	0.940	0.929
Cos.	Uniform	10	Het.	0.121	0.117	3.7	0.957	0.929
Cos.	Trunc. normal	5	Homo	0.156	0.153	2.1	0.933	0.927
Cos.	Trunc. normal	10	Homo	0.107	0.106	0.8	0.947	0.928
Cos.	Trunc. normal	5	Het.	0.182	0.179	1.8	0.960	0.926
Cos.	Trunc. normal	10	Het.	0.125	0.119	5.2	0.957	0.929

Notes: Each design uses 300 replications. Sep. is the equal-weight average of cutoff-specific local linear RD estimates. Comb. uses the feasible covariance-aware convex combination for each segment. Gain is its percentage RMSE reduction relative to Sep.; Cov.95 is its empirical Wald coverage.

Table 2: Multi-threshold simulation results

The covariance-aware decomposition estimator reduces RMSE relative to the matched-bandwidth separate-RD benchmark in all 16 designs. The median gain is 2.9%, with a range from 0.6% to 5.2%. Coverage ranges from 93.3% to 96.3%. The diagonal rule improves in 15 designs, whereas the covariance-aware rule also improves in the remaining design; the difference between the two combinations is otherwise small because the average segment-level correlation is about 0.08. These comparisons hold the local bandwidth rule fixed across estimators and do not claim dominance under separate estimator-specific optimal tuning.

7 Discussion and Conclusion

This paper studies estimation of smooth segment contrasts that arise in ordered multi-threshold regression discontinuity designs. We show that aggregate RD estimands can be decomposed into two components: local discontinuities at thresholds and smooth between-threshold contrasts. While standard

RD methods identify the former using observations near the cutoffs, the latter depend on differences of endpoint limits across adjacent segments, such as $m(1^-) - m(0^+)$ after normalization.

The main contribution is to provide a second estimator for these smooth segment contrasts. The integrated-derivative estimator exploits the identity

$$m(1^-) - m(0^+) = \int_0^1 m'(u) du,$$

and uses interior observations to estimate the derivative and aggregate it over the segment. The estimator admits a feasible central limit theorem with an explicit variance expression that depends on the design density and conditional variance over the segment.

A key result is that the correlation between the integrated-derivative and boundary-level estimators vanishes. Their covariance is nevertheless retained in the recommended minimum-variance combination because the potential improvement over boundary estimation is itself a higher-order effect. Under regular designs the interior estimator has variance of order $(Nh^2)^{-1}$, compared with $(Nh)^{-1}$ for the boundary estimator, so the combined estimator is first-order equivalent to the boundary estimator. The paper therefore claims finite-sample augmentation, not a first-order efficiency improvement.

The implications are design-dependent. At the matched bandwidth, the covariance-aware simulations show modest gains in every reported design, with larger improvements when endpoint information is relatively weak. Because the integrated estimator converges more slowly under regular designs, these are finite-sample gains rather than a first-order efficiency improvement. The contribution is therefore a feasible augmentation and combination rule, not a claim of uniform dominance over every separately tuned endpoint procedure.

The approach relies on smoothness of the conditional mean within each segment. If there are latent discontinuities or structural breaks between observed thresholds, integrating the ordinary derivative omits those jumps and

the estimator is inconsistent for the endpoint contrast. Finite-sample performance also depends on bandwidth choices, quadrature, and local sample sizes. These restrictions should be made empirically visible through specification checks and sensitivity analysis rather than treated as innocuous regularity conditions.

Finally, the analysis assumes known and ordered thresholds. Extensions to settings with endogenous or estimated cutoffs, as well as to designs with multidimensional running variables or interacting thresholds, remain open. More broadly, the decomposition in (1) separates boundary discontinuities from interior smooth variation and requires that any combination respect their different identification content.

Deriving a joint bandwidth selector, robust bias correction, and extensions to nonregular endpoint densities are useful directions for future research.

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Appendix

A Proofs

A.1 Proof of Theorem 1

Write

$$\hat{\beta}_{\text{avg}} - \Gamma = 2h \sum_{b=1}^B (\hat{\beta}_1(c_b) - m'(c_b)) + \left\{ 2h \sum_{b=1}^B m'(c_b) - \int_0^1 m'(u) du \right\}.$$

By Lemma 3, the conditional expectation of the first term is

$$2h \sum_{b=1}^B (\mathbb{E}[\hat{\beta}_1(c_b) \mid X] - m'(c_b)) = \frac{h^p}{(p+1)!} \left(\int_{-1}^1 u^{p+1} K^*(u) du \right) [m^{(p)}(1^-) - m^{(p)}(0^+)] + o(h^p).$$

By Lemma 2, applied to $g = m'$,

$$2h \sum_{b=1}^B m'(c_b) - \int_0^1 m'(u) du = -\frac{h^2}{6} [m''(1^-) - m''(0^+)] + O(h^4).$$

Thus, if $p > 2$, the leading bias is

$$\text{Bias} = -\frac{h^2}{6} [m''(1^-) - m''(0^+)].$$

If $p = 2$, the local-polynomial and midpoint terms both contribute at order h^2 , giving

$$\text{Bias} = \frac{h^2}{6} \left(\int_{-1}^1 u^3 K^*(u) du - 1 \right) [m''(1^-) - m''(0^+)].$$

For the stochastic term, write

$$\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X] = \sum_{i=1}^N w_{N,i} \varepsilon_i.$$

By Lemma 4,

$$\text{Var} \left(\sum_{i=1}^N w_{N,i} \varepsilon_i \mid X \right) = \frac{2}{Nh^2} \left(\int_{-1}^1 K^{*2}(u) du \right) \left(\int_0^1 \frac{\sigma^2(u)}{f(u)} du \right) + o \left(\frac{1}{Nh^2} \right).$$

Let

$$\Omega^2 = 2 \left(\int_{-1}^1 K^{*2}(u) du \right) \left(\int_0^1 \frac{\sigma^2(u)}{f(u)} du \right).$$

Lemma 1 gives

$$\frac{\sqrt{Nh^2}}{\Omega} (\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X]) \xrightarrow{d} \mathcal{N}(0, 1).$$

The remaining deterministic terms are negligible under Assumption 5. Combining the bias expansion, the conditional CLT, and Slutsky's lemma proves the result. $\square$

A.2 Proof of Theorem 2

Write

$$S_\beta = \hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X] = \sum_{i=1}^N w_i \varepsilon_i$$

and

$$S_\delta = \hat{\delta}_{\text{levels}} - \mathbb{E}[\hat{\delta}_{\text{levels}} \mid X] = \sum_{i=1}^N v_i \varepsilon_i.$$

Conditional on X ,

$$\text{Cov}(S_\beta, S_\delta \mid X) = \sum_{i=1}^N w_i v_i \sigma^2(X_i).$$

By construction, $v_i \neq 0$ only if

$$X_i \in \mathcal{O}_h := [0, A_0 h] \cup [1 - A_1 h, 1].$$

Since f has finite one-sided limits at the endpoints and no atoms,

$$\mathbb{P}(X_i \in \mathcal{O}_h) = A_0 h f(0+) + A_1 h f(1-) + o(h).$$

Hence,

$$\#\{i : X_i \in \mathcal{O}_h\} = N\{A_0f(0+) + A_1f(1-)\}h + o_p(Nh).$$

Using

$$\max_i |w_i| \leq \frac{C}{Nh}, \quad \max_i |v_i| \leq \frac{C}{Nh},$$

and boundedness of $\sigma^2(\cdot)$,

$$|\text{Cov}(S_\beta, S_\delta \mid X)| \leq C\#\{i : X_i \in \mathcal{O}_h\} \frac{1}{Nh} \frac{1}{Nh} = O_p\left(\frac{A_0f(0+) + A_1f(1-)}{Nh}\right).$$

Moreover,

$$\text{Var}(S_\beta \mid X) = \Theta_p\left(\frac{1}{Nh^2}\right), \quad \text{Var}(S_\delta \mid X) = \Theta_p\left(\frac{1}{Nh}\right).$$

Therefore,

$$\text{Corr}(S_\beta, S_\delta \mid X) = O_p\left(\{A_0f(0+) + A_1f(1-)\}h^{1/2}\right) \rightarrow 0.$$

For joint convergence, fix constants a and b and form the corresponding linear combination of the standardized statistics. It is again a triangular array of independent, conditionally mean-zero terms. The same maximum-weight and $(2 + \eta)$ -moment bounds used in Lemma 1 verify its Lyapunov condition. Its conditional variance converges to $a^2 + b^2$ because the standardized covariance tends to zero. The Cramér–Wold device therefore gives joint asymptotic normality with diagonal limiting covariance. $\square$

Remark 2 (Quadrature order and bandwidth restrictions). The midpoint aggregation induces an $O(h^2)$ quadrature error for $\int_0^1 m'(u) du$. If a composite quadrature rule of global order $q \geq 2$ is used instead, the quadrature error becomes $O(h^q)$. Since the aggregate local-polynomial bias is $O(h^p)$, the deterministic remainder is $O(h^{\min\{p, q\}})$. Under the normalization $\sqrt{Nh^2}$, it suffices that

$$\sqrt{Nh}^{1+\min\{p, q\}} \rightarrow 0.$$

Thus, the bandwidth restriction in Theorem 1 can be relaxed with higher-order quadrature rules.

A.3 Auxiliary lemmas

Lemma 1 (Convergence in Distribution). *Under Assumptions 1–6, for each sample size N , consider the triangular array*

$$\{w_{N,i}\varepsilon_i : i = 1, \dots, N\}.$$

Define

$$T_N = \sum_{i=1}^N w_{N,i}\varepsilon_i, \quad v_N^2 = \text{Var}(T_N \mid X) = \sum_{i=1}^N w_{N,i}^2 \sigma^2(X_i).$$

Suppose there exists a deterministic sequence $\alpha_N > 0$ such that

$$\alpha_N^2 v_N^2 \xrightarrow{p} \Omega^2 > 0.$$

Then

$$\frac{\alpha_N T_N}{\Omega} \xRightarrow{d} \mathcal{N}(0, 1).$$

Proof. The effective weights satisfy

$$\max_i |w_{N,i}| = O_p\left(\frac{1}{Nh}\right), \quad \sum_{i=1}^N w_{N,i}^2 = \Theta_p\left(\frac{1}{Nh^2}\right).$$

Conditional on X , the summands $w_{N,i}\varepsilon_i$ are independent and mean zero. By Assumption 6,

$$\sum_{i=1}^N \mathbb{E} \left[|w_{N,i}\varepsilon_i|^{2+\eta} \mid X \right] \leq C \sum_{i=1}^N |w_{N,i}|^{2+\eta} = O_p\left(\frac{1}{N^{1+\eta}h^{2+\eta}}\right).$$

Moreover, Assumption 3 implies $v_N^2 \asymp_p \sum_i w_{N,i}^2$, and hence

$$(v_N^2)^{1+\eta/2} = \Theta_p\left(\frac{1}{N^{1+\eta/2}h^{2+\eta}}\right).$$

Therefore the Lyapunov ratio is $O_p(N^{-\eta/2}) = o_p(1)$. The triangular-array Lyapunov CLT gives

$$\frac{T_N}{v_N} \xrightarrow{d} \mathcal{N}(0, 1).$$

Since $\alpha_N^2 v_N^2 \rightarrow_p \Omega^2$, Slutsky's lemma proves the result. $\square$

Lemma 2 (Midpoint Rule Approximation Error). *Let $g \in C^4([0, 1])$. Partition $[0, 1]$ into B subintervals of length $2h = 1/B$, with midpoints $c_b = (2b - 1)h$. Define*

$$E = \int_0^1 g(u) du - 2h \sum_{b=1}^B g(c_b).$$

Then

$$E = \frac{h^2}{6} \{g'(1^-) - g'(0^+)\} + O(h^4).$$

Proof. On each interval $[a_b, a_b + 2h]$, with midpoint $c_b = a_b + h$, Taylor expansion gives

$$\int_{a_b}^{a_b+2h} g(u) du = 2h g(c_b) + \frac{1}{3} h^3 g''(c_b) + O(h^5).$$

Summing over b yields

$$E = \frac{1}{3} h^3 \sum_{b=1}^B g''(c_b) + O(h^4).$$

Since

$$2h \sum_{b=1}^B g''(c_b) = \int_0^1 g''(u) du + O(h^2) = g'(1^-) - g'(0^+) + O(h^2),$$

the result follows. $\square$

Lemma 3 (Aggregate Bias of Local Polynomial Estimator). *Under Assumptions 1–6, let $\hat{\beta}_1(c_b)$ denote the degree- p local-polynomial estimator of $m'(c_b)$. Partition $[0, 1]$ into B subintervals of width $2h = 1/B$, with midpoints $c_b = (2b - 1)h$, and define*

$$R = 2h \sum_{b=1}^B \hat{\beta}_1(c_b) - 2h \sum_{b=1}^B m'(c_b).$$

Let $\mathcal{B}_N(h) = \mathbb{E}[R \mid X]$. Then

$$\mathcal{B}_N(h) = \frac{h^p}{(p+1)!} \left(\int_{-1}^1 u^{p+1} K^*(u) du \right) \{m^{(p)}(1^-) - m^{(p)}(0^+)\} + o(h^p).$$

Proof. For each interior midpoint c_b , the standard equivalent-kernel expansion gives

$$\mathbb{E}[\hat{\beta}_1(c_b) \mid X] - m'(c_b) = \frac{h^p}{(p+1)!} m^{(p+1)}(c_b) \int_{-1}^1 u^{p+1} K^*(u) du + h^p \rho_b(h),$$

where $\sup_b |\rho_b(h)| = o(1)$. Multiplying by $2h$ and summing gives

$$\mathcal{B}_N(h) = \frac{2h^{p+1}}{(p+1)!} \left(\int_{-1}^1 u^{p+1} K^*(u) du \right) \sum_{b=1}^B m^{(p+1)}(c_b) + o(h^p).$$

Since

$$2h \sum_{b=1}^B m^{(p+1)}(c_b) \rightarrow \int_0^1 m^{(p+1)}(u) du = m^{(p)}(1^-) - m^{(p)}(0^+),$$

the result follows. $\square$

Lemma 4 (Variance of the Integrated-Derivative Estimator). *Under Assumptions 1–6,*

$$\text{Var}(\hat{\beta}_{\text{avg}} \mid X) = \frac{2}{Nh^2} \int_{-1}^1 K^{*2}(u) du \int_0^1 \frac{\sigma^2(u)}{f(u)} du + o\left(\frac{1}{Nh^2}\right).$$

Proof. Because the kernel is supported on $[-1, 1]$ and the bins have width $2h$, the observations used in different midpoint regressions are disjoint. Conditional on X ,

$$\text{Var}(\hat{\beta}_{\text{avg}} \mid X) = 4h^2 \sum_{b=1}^B \text{Var}(\hat{\beta}_1(c_b) \mid X).$$

The local-polynomial variance expansion gives, uniformly over b ,

$$\text{Var}(\hat{\beta}_1(c_b) \mid X) = \frac{\sigma^2(c_b)}{Nh^3 f(c_b)} \int_{-1}^1 K^{*2}(u) du + o\left(\frac{1}{Nh^3}\right).$$

Therefore,

$$\text{Var}(\hat{\beta}_{\text{avg}} \mid X) = \frac{4}{Nh} \left(\int_{-1}^1 K^{*2}(u) du \right) \sum_{b=1}^B \frac{\sigma^2(c_b)}{f(c_b)} + o \left(\frac{1}{Nh^2} \right).$$

Since

$$2h \sum_{b=1}^B \frac{\sigma^2(c_b)}{f(c_b)} \rightarrow \int_0^1 \frac{\sigma^2(u)}{f(u)} du,$$

the result follows. □